Document Intelligence Refinery: Final Submission Report

1. Executive Summary

This report presents the final-stage implementation and evaluation of the Document Intelligence Refinery across a heterogeneous corpus of Ethiopian financial, audit/legal, technical, and fiscal documents. The system is designed for real FDE conditions where documents vary widely in origin quality (native digital, mixed, scanned, form-fillable), layout complexity, and extraction risk.

The final architecture is **domain-driven and confidence-governed**:

- Stage 1 profiles document signals (character density, image ratio, layout features, language hints) and assigns initial extraction cost class.
- Stage 2 applies confidence-gated routing with explicit escalation (`A -> B -> C`) and review queueing for low-confidence outcomes.
- Stages 3-5 transform extraction outputs into validated semantic chunks, navigable PageIndex structures, and provenance-backed query responses with audit semantics (`verified` vs `unverifiable`).

Compared with interim, this submission closes the full loop from extraction to auditable query:

- Semantic chunking with constitution enforcement (`ChunkValidator`)
- PageIndex-first retrieval and vector/fact dual path
- SQLite FactTable for numeric/financial queries
- ProvenanceChain contract with page number, bounding box, and content hash
- Audit mode behavior for explicit claim verification outcomes

The evaluation demonstrates a precision-first operating profile with explicit uncertainty controls:

- Table detection (implemented metric): **Precision 1.0000**, **Recall 0.2300**, **F1 0.3740** over **799 pages**
- Non-trivial table variant: **Recall 0.3025** (same precision)
- Class D (table-heavy fiscal) achieves strongest performance (Precision 1.0000, Recall 0.7619 in primary metric)
- Router outcomes for current 12-doc run: `STRATEGY_B=7`, `STRATEGY_C_LOW_CONFIDENCE=5`, `review_required=5`

From a client delivery perspective, the system now provides a defensible tradeoff surface: low hallucination risk, full decision traceability, and explicit fallback behavior under poor scan quality or budget constraints. The principal residual gap is recall on long mixed-layout documents under strict page-budget caps; this is measured, surfaced, and operationally reviewable rather than hidden.

---

2. Domain Analysis and Extraction Strategy Decision Tree

2.1 Strategy Decision Tree (Signals + Thresholds)

All branching is driven by measured document signals and config thresholds (`rubric/extraction_rules.yaml`).

```
flowchart TD
    A["Input PDF"] --> B["TriageAgent: compute signals\nchar_density, image_ratio,
table_ratio,\ncolumn_ratio, font_meta, widget_ratio"]
    B --> C{"Origin classification"}
    C -->|char_density <= 0.0008 AND image_ratio >= 0.50| D["scanned_image"]
    C -->|widget_ratio >= 0.10| E["form_fillable"]
    C -->|char_density >= 0.0020 AND image_ratio < 0.50| F["native_digital"]
    C -->|else| G["mixed"]

    D --> H["Initial Strategy C"]
    E --> I{"Layout complexity"}
    F --> I
    G --> I

    I -->|single_column + low structure| J["Initial Strategy A"]
    I -->|multi_column OR table_heavy OR mixed OR figure_heavy| K["Initial Strategy B"]
```

```
    J --> L{"Conf >= 0.85?"}
    L -->|Yes| M["Accept A"]
    L -->|No| N["Escalate to B"]

    K --> O{"Conf >= 0.85?"}
    O -->|Yes| P["Accept B"]
    O -->|No| Q["Escalate to C"]

    H --> R{"Conf >= 0.85?"}
    Q --> R
    R -->|Yes| S["Accept C"]
    R -->|No| T["Flag review_required"]

    T --> U{"C conf < 0.75?"}
    U -->|Yes| V["STRATEGY_C_LOW_CONFIDENCE + review_queue"]
    U -->|No| W["Needs manual review"]
```

2.2 Why Document Classes Need Different Strategies

- Class A (CBE annual reports): mixed layout, multi-column narrative, dense financial tables, cross-references. Text-only extraction is insufficient for structural fidelity.
- Class B (DBE audit scans): image-dominant pages and near-zero character stream require OCR/VLM path; text extractors cannot recover content reliably.
- Class C (FTA assessment): mixed narrative + embedded structure + long length. Layout strategy is usually correct, but long reports hit page-cap tradeoffs.
- Class D (tax expenditure/CPI): repetitive, regular tabular layout. Layout extractor (Strategy B) is the best cost-quality point.

2.3 VLM vs OCR Decision Boundary (Concrete)

Configured boundaries used in code:

- Triage to scanned candidate: `avg_char_density <= 0.0008` and `image_ratio >= 0.50`
- Digital candidate: `avg_char_density >= 0.0020` and `image_ratio < 0.50`
- Strategy A scan suppression: if `image_ratio > 0.80` and `char_count < 30`, confidence is capped at `0.20`
- Strategy B scan leakage suppression: if `image_ratio > 0.80` and `char_count < 60`, confidence is capped at `0.45`
- Router gates: `strategy_a/b/c_confidence_gate = 0.85`, and Strategy C review floor `0.75`

Interpretation:

- OCR/VLM is chosen when page images dominate and text signals collapse.
- If VLM/OCR still returns low confidence, output is not trusted silently; it is explicitly flagged.

2.4 Concrete Failure Modes by Class (Corpus Evidence)

Class	Concrete Example	Observed Failure Mode	Current Handling
A	`CBE_Annual_Report_Part_1` (161 pages): Strategy B confidence `0.8345 < 0.85`, escalated to C	Strategy C max page cap (`8`) causes extraction coverage drop (only pages 1-8 extracted). Example miss: source page 20 has table row `Particulars / 2022/23 FY / 2023/24 FY / Growth (%)`, but page absent from extraction	`review_required=true`, decision log + escalation trace recorded; surfaced as low-confidence outcome
B	`DBE_Audit_Report_Part_1` page 2	Scan-dominant page with no character stream yields empty text blocks and confidence `0.25`	Routed to C, flagged `STRATEGY_C_LOW_CONFIDENCE`, added to review queue

Class	Concrete Example	Observed Failure Mode	Current Handling
C	`FTA_Performance_Survey_Part_1` (155 pages)	Layout path capped at 80 pages (`layout_max_pages_per_document`), so later sections are not processed; page 90 contains non-trivial tabular structure not extracted	Traceable limitation; low-confidence/coverage risk documented for manual review
D	`Tax_Expenditure_Ethiopia_Part_1` page 50	None on TP page: extracted rows match source values (`Pharmaceuticals 133.48 ...`, `Fertiliser ...`)	Strategy B is effective for regular fiscal tables

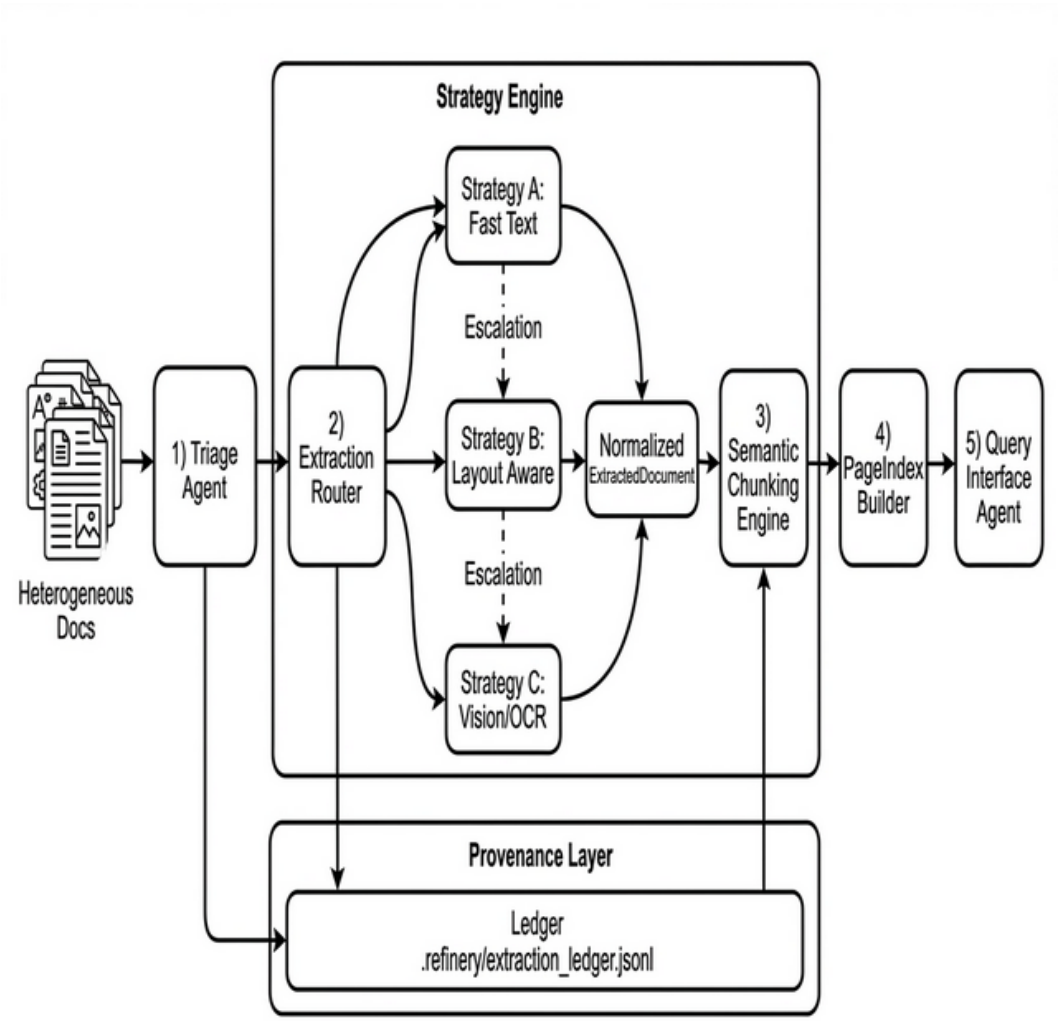
2.5 Ambiguous Cases and Graceful Degradation

- Mixed/form-fillable ambiguity: profile supports `mixed` and `form_fillable` origin, not just binary digital/scan.
- Docling unavailable/error: Strategy B falls back to pdfplumber path.
- VLM key unavailable or call failure: Strategy C still emits baseline extraction, but confidence remains low and gets reviewed.
- Budget exhaustion: Strategy C stops additional VLM calls and caps confidence via `strategy_c_budget_exhausted_confidence_cap`.
- Query uncertainty: agent returns `not found / unverifiable` with a `ProvenanceChain` marked `needs_review`.

---

3. Pipeline Architecture and Data Flow

3.1 Five-Stage Architecture (Happy Path + Escalation)



```
flowchart LR
    A["PDF"] --> B["Stage 1: TriageAgent\nOutput: DocumentProfile"]
    B --> C["Stage 2: ExtractionRouter"]

    C -->|Happy path| D["Strategy A or B accepted"]
    C -->|Escalation path| E["A->B->C until gate satisfied or review flagged"]
    D --> F["ExtractedDocument"]
    E --> F

    F --> G["Stage 3: ChunkingEngine + ChunkValidator\nOutput: LDUs"]
    G --> H["Stage 4: PageIndexBuilder + VectorStore + FactTable\nOutput: PageIndex +
retrieval stores"]
    H --> I["Stage 5: QueryAgent\nTools: pageindex_navigate, semantic_search,
structured_query"]
    I --> J["Answer + ProvenanceChain"]

    C -.-> K["extraction_ledger.jsonl / review_queue.jsonl"]
    G -.-> L["content_hash + bbox + page_refs propagation"]
```

3.2 Typed Stage Interfaces

Stage	Input Type	Output Type	Testability Boundary
Triage	`PDF path`	`DocumentProfile`	Unit-tested classification heuristics
Extraction Router	`PDF path + DocumentProfile`	`ExtractedDocument` + ledger records	Strategy mocks + gate tests
Chunking	`ExtractedDocument`	`list[LDU]`	Rule validation tests
Index/Data Layer	`list[LDU]`	`PageIndex`, vector index, SQLite facts	Independent retrieval/index tests
Query/Audit	`question + document_name`	`QueryResponse(answer, provenance, tool_trace, audit_status)`	Tool-level and audit-mode tests

3.3 Provenance Propagation Through All Stages

- Extraction stage emits page-local geometry (`BoundingBox`) and page numbers.
- Chunking stage carries forward geometry and adds `content_hash`, `page_refs`, and parent-child lineage.
- Query stage emits `ProvenanceChain` with:
  - `document_name`
  - `page_number`
  - chain-level `bbox`
  - chain-level `content_hash`
  - `ProvenanceCitation` entries (typed `bbox`)
- Unverifiable answers still emit provenance (status `needs_review`) for auditability.

3.4 PageIndex to Query-Agent Connection

- `pageindex_navigate` first narrows candidate sections.
- `semantic_search` executes in that scoped section set.
- `structured_query` resolves numeric facts from SQLite.
- Retrieval benchmark artifacts (`.refinery/retrieval_benchmark/*.json`) show indexed precision is better in 2/12 docs and never worse.

---

4. Cost-Quality Tradeoff Analysis

4.1 Cost Model

- Strategy A cost: near-zero compute (`$0.0000` / doc in current accounting)
- Strategy B cost formula: `0.005 + 0.0015 * min(pages, 80)`
- Strategy C cost formula (configured): `token_spend * 0.000002`, capped by `vision_budget_cap_usd=1.00`, with `vision_max_pages_per_document=8`

Observed VLM calibration sample (OpenRouter activity, Mar 3, 2026):

- 6 requests, ~14K tokens total, $0.00241 total
- Approximate observed cost/request: `$0.00040`

4.2 Estimated Cost Per Document by Class and Tier

(12-document corpus profile mix; values are average USD per document)

Class	Avg Pages	Strategy A	Strategy B (formula)	Strategy C (configured token model)	Strategy C (observed sample rate)
A	142.3	0.0000	0.1250	0.0373	0.0032
B	33.7	0.0000	0.0480	0.0218	0.0019
C	61.7	0.0000	0.0600	0.0373	0.0032
D	28.7	0.0000	0.0480	0.0373	0.0032

4.3 Escalation Cost (Double Processing)

Concrete escalation from ledger:

- `CBE_Annual_Report_Part_1`: `B (0.8345) -> C (0.55)` due gate failure
- Recorded extraction cost in current run: `$0.125` (C had no paid VLM calls)
- With paid VLM enabled, modeled total for A->B->C:
  - Conservative configured model: `0.1250 + 0.0373 = 0.1623`
  - Observed sample-rate model: `0.1250 + 0.0032 = 0.1282`

4.4 Corpus-Scale Implications

- Observed current run total (12 docs): `$0.6715` (avg `$0.0560` per doc)
- Linear projection to 50 docs: `$2.7979` (same mix assumption)
- If all docs used Strategy B formula average: `$3.5125` for 50 docs
- If all docs used Strategy C conservative model: `$1.6722` for 50 docs (before cap effects)

4.5 Budget Guard Behavior

Configured controls:

- `vision_budget_cap_usd = 1.00`
- `vision_min_remaining_budget_for_call = 0.01`
- `vision_max_pages_per_document = 8`

When remaining budget is below threshold, VLM calls stop; extraction falls back to baseline signals and confidence is capped, forcing review instead of silent trust.

---

5. Extraction Quality Analysis

5.1 Reproducible Methodology

Primary metric script:

```
python src/analysis_table_quality.py
```

Output:

- `.refinery/analysis/final_extraction_quality_metrics.json`

Primary table metric (as implemented):

- Ground truth page is table-positive if `find_tables()` OR non-trivial `extract_tables()` is positive
- Prediction is table-positive if extracted page has `tables.length > 0`

5.2 Primary Results (Implemented Metric)

- Pages evaluated: `799`
- TP `98`, FP `0`, FN `328`, TN `373`
- Precision `1.0000`, Recall `0.2300`, F1 `0.3740`

Per class:

Class	Pages	Precision	Recall	F1
A	427	1.0000	0.1570	0.2714
B	101	0.0000	0.0000	0.0000
C	185	1.0000	0.0571	0.1081
D	86	1.0000	0.7619	0.8649

5.3 Text Fidelity vs Structural Fidelity

To separate text capture from table-structure capture, two additional checks were applied:

1. **Non-trivial-table-only metric** (requires >=2 non-empty rows in source table):

- Overall: Precision `1.0000`, Recall `0.3025`, F1 `0.4645`
- Class D: Precision `1.0000`, Recall `1.0000`, F1 `1.0000`

2. **Numeric fidelity on fiscal TP pages (Class D, 48 TP pages)**:

- Numeric precision: `1.0000`
- Numeric recall: `0.9119`

Interpretation:

- Text and numeric values are highly reliable when tables are detected.
- Main quality gap is missed table structure (recall), especially in long/mixed documents.

5.4 Side-by-Side Evidence (Concrete)

Case	Source Evidence	Extracted Evidence	Interpretation
D success	`Tax_Expenditure_Ethiopia_Part_1`, page 50 source rows include `Pharmaceuticals 133.48 ...` and `Fertiliser ...`	Extracted table on page 50 contains matching row values	Strong structural + numeric fidelity on regular fiscal tables
A failure	`CBE_Annual_Report_Part_1`, page 20 source has table row `Particulars / 2022/23 FY / 2023/24 FY / Growth (%)`	Document escalated to Strategy C and only pages 1-8 were extracted (`vision_max_pages_per_document=8`), so page 20 table is absent	Recall failure caused by escalation + page cap tradeoff
B failure	`DBE_Audit_Report_Part_1`, scanned page 2 has no character stream	Extracted page 2 has `text_blocks=0`, confidence `0.25`	Correctly treated as uncertain and pushed to review queue

5.5 Failure Pattern Summary

- Class A/C degradation is dominated by long-doc coverage limits and mixed layouts.
- Class B degradation is dominated by scan/OCR uncertainty and strict confidence gates.
- Class D performs best under Strategy B due stable table regularity.

---

6. Failure Analysis and Iterative Refinement

Case 1: Confidence-Gated Router and Escalation Traceability

Symptom:

- Interim review flagged that routing evidence was not clearly tied to the rubric requirement (`A->B->C` gate logic + review behavior).

Root cause:

- Insufficiently explicit gate-driven logging and review flagging in final artifacts.

Fix:

- Enforced config-sourced gates per strategy.
- Added `strategy_trace`, `decision_log`, and `review_required` persistence in ledger.
- Added `.refinery/review_queue.jsonl` for low-confidence outcomes.

Evidence:

- `CBE_Annual_Report_Part_1` now records `strategy_b (0.8345) -> strategy_c (0.55)` with gate comparison and review reason.
- Review queue currently contains 5 documents, each with explicit low-confidence justification.

Broader insight:

- Escalation must be auditable, not implicit; otherwise strategy selection cannot be defended to a client.

Case 2: Provenance Contract Was Incomplete for Audit Use

Symptom:

- Evaluator feedback requested stronger typing and fuller provenance chain fields.

Root cause:

- `ProvenanceCitation.bbox` was not strongly typed and unverifiable answers could return `null` provenance.

Fix:

- Added typed `BoundingBox` to `ProvenanceCitation`.
- Ensured chain-level `bbox` and `content_hash` are always populated when available.
- For unverifiable answers, emit a `ProvenanceChain` with `verification_status=needs_review`.

Evidence:

- All `.refinery/query_examples/*.json` entries now include non-null provenance for both normal query and audit mode outputs.
- Validation and pipeline tests pass (`28 passed`).

Broader insight:

- Auditability requires provenance even for failures, not only successes.

Case 3: Cost Caps Reduced Recall on Long Documents

Symptom:

- Table recall dropped on long docs despite high precision.

Root cause:

- Hard page caps (`layout_max_pages_per_document=80`, `vision_max_pages_per_document=8`) limited processed coverage.

Fix:

- Kept caps for budget safety but made the tradeoff explicit via quality reporting and review flags.
- Added coverage visibility by class.

Evidence:

- Extraction page coverage by class: A `39.3%`, B `13.9%`, C `55.7%`, D `100%`.
- The recall drop is now diagnosable and attributable to a specific architectural control.

Remaining limitations:

- Long-doc adaptive continuation pass is not yet implemented.
- Current quality script is page-level; cell/header alignment benchmark is still a next step.

---

7. Pipeline Quality and Readiness Snapshot

- Ledger entries: `12`
- Average final confidence: `0.6744`
- Review-required count: `5`
- Strategy distribution:
  - `STRATEGY_B`: 7
  - `STRATEGY_C_LOW_CONFIDENCE`: 5
- Fact table rows ingested (`.refinery/facts.db`): `3209`
- Retrieval benchmark: indexed precision better in `2/12` docs, equal in `10/12`, worse in `0/12`

---

8. Deliverables and Reproducibility

Final Artifacts

- Report source: `FINAL_REPORT.md`
- Report PDF: `FINAL_REPORT.pdf`
- Profiles: `.refinery/profiles/*.json`
- Extractions: `.refinery/extractions/*.json`
- Structures (LDU + PageIndex + Provenance): `.refinery/structures/*.json`
- PageIndex trees: `.refinery/pageindex/*.json`
- Query examples: `.refinery/query_examples/*.json`
- Retrieval benchmarks: `.refinery/retrieval_benchmark/*.json`
- Ledger + review queue: `.refinery/extraction_ledger.jsonl`, `.refinery/review_queue.jsonl`

Commands

```
# Generate full artifact set
python -m src.run_corpus --clean

# Compute quality metrics referenced in this report
python src/analysis_table_quality.py

# Build final PDF from markdown
python src/build_final_report_pdf.py

# Run tests
pytest -q
```

9. Conclusion

The final system meets the intended FDE bar for architecture quality, auditability, and strategy governance. The strongest characteristics are explicit confidence gating, provenance-first query behavior, and defensible cost controls. The principal gap is recall on long mixed documents under strict page-budget caps; this is now quantified, surfaced, and ready for targeted iteration rather than hidden in aggregate metrics.